

Department of Computer Science & Engineering

Subject Code:	CSL48	TERM: -Feb-June 2026	
Subject Name:	System Programming Laboratory	Faculty In-charge:	CP/SV/DN/VCD
Credits:	0:0:1	Semester :	IV

Exercise Number	Problem Statements	CO	PO
1.	a) Write a program to demonstrate how fork() creates separate parent and child processes with independent copies of global and local variables after process creation b) Write a program to demonstrate Inter-Process Communication (IPC) using shared memory by implementing the Producer–Consumer problem.	CO1	PO 1-3
2.	a) Write a C program to demonstrate Two level file organization technique. b) Write a program to demonstrate synchronization in shared resource access using the Readers–Writers problem.	CO1	PO 1-3
3.	a) Write a C program to implement Non Preemptive SJF CPU Scheduling b) Write a C program to simulate Bankers Algorithm for Deadlock Avoidance	CO1	PO 1-3
4.	a) Write a C program to implement Round Robin CPU Scheduling b) Write a program to implement the contiguous memory strategies algorithms: First Fit, Best Fit, and Worst Fit for contiguous allocation.	CO1 CO2	PO 1-3
5.	a) Write a program to implement Shortest Remaining Time First (SRTF) CPU Scheduling b) Write a program to implement Page replacement algorithms: FIFO, LRU and Optimal	CO2 CO3	PO 1-3
6.	a) Write a C program to demonstrate Single level file organization techniques b) Implementation of Disk Scheduling using FCFS and SCAN algorithms.	CO1 CO2	PO 1-3
7.	a) Write a C program to implement Non Preemptive FCFS CPU Scheduling	CO2	PO 1-3

	b) Write a C program to demonstrate synchronization and deadlock handling using the Dining Philosophers problem.		
8.	a) Write a c program to demonstrate synchronization between producer-consumer processes that share a common buffer and performs a read and write operation to and from the buffer. b) Write a C program to simulate memory and CPU virtualization	CO1 CO2	PO 1-3
9.	a) Write a C program to simulate the Paging and Segmentation techniques. b) Write a program to demonstrate the use of the exec() system call (execvp) to replace the child process with a new program, showing how a process executes an external command after fork().	CO1	PO 1-3
10.	a) Write a C program to demonstrate Inter-Process Communication (IPC) using message passing by implementing the Producer–Consumer problem. b) Write a C program to demonstrate file allocation using the Contiguous allocation approaches	CO3 CO2	PO 1-3
11.	a) Write a C program to demonstrate file allocation using the Linked file allocation approach. b) Write a program to perform the following operations i. To create a child process ii. The child process should execute a separate program (using exec function) that calculates the addition of two numbers by passing two integer values. iii. The parent process should wait for a child to complete	CO1 CO3	PO 1-3
12.	a) Write a program to implement Preemptive Priority CPU Scheduling. b) Write a C program to demonstrate file allocation using the Indexed allocation approach	CO2	PO 1-3

Marks Distribution

Conduction and Result	Write-up	Execution	Viva	Change of Program
	7M	Part a 7M Part b 8M	3M	-3M

Course Coordinator

Reviewer

HOD

HOD, Dept. of CSE